

Project Rosetta: The Approximation Entropy & The Fractal Limits of Digital Physics

Quantifying the Irreducible Cost of Translating Continuous
Physical Dynamics into Discrete Digital Computation

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Abstract

Digital computation has been elevated from an engineering convenience to an ontological assumption – the belief that physical reality is, at bottom, computable in binary. We challenge this assumption. Through a systematic metamathematical analysis connecting symplectic geometry, transcendental function theory, quantum thermodynamics, and complexity theory, we prove that translating continuous bosonic dynamics into discrete digital logic carries an irreducible thermodynamic cost – **Approximation Entropy** S_A . For the superconducting transmon qubit, we derive $S_A \approx k_B T \ln(1/\alpha_r)$ and define **Rosetta’s Constant** $R = k_B T \ln(1/\alpha_r) \approx 8.2 \times 10^{-25}$ Joules per logical translation at 15 mK. We prove that the translation functor $F : \mathcal{P} \rightarrow \mathcal{C}$ from continuous physical dynamics to finite gate circuits is **non-exact**. We identify the **Hidden Axioms Layer** – Anthropocentricity, Radix Contingency, and the Archimedean Continuum Fallacy – and propose a falsifiable experimental protocol: the **Translation Calorimeter**, a cryogenic measurement of the differential heat between analog evolution and digital gate translation on a single transmon.

1. Introduction

1.1 The Map-Territory Problem

The global quantum computing industry has absorbed approximately 35 billion dollars in combined public and private investment over two decades while delivering zero commercially viable machines (QNFO Research Collective 2026e). This paper argues that the failure is not primarily an engineering problem but a **category error** – the systematic confusion of a mathematical map (the qubit-gate-circuit model) with the physical territory (continuous bosonic dynamics).

The qubit is a **scaffold**, not an invariant. The gate is a projection, not an operation. The surface code is a mathematical trick, not a physical law.

1.2 Statement of Results

Axis	Core Result
1. Categorical Foundations	The functor $F : \mathcal{P} \rightarrow \mathcal{C}$ is non-exact (Theorem 1)
2. Algebraization Complexity	$\cos(\phi)$ requires degree-224 polynomial for spectral fidelity 10^{-4}
3. Thermodynamics of Translation	Rosetta’s Constant $R = k_B T \ln(1/\alpha_r)$
4. Complexity Bounds	Digital Speed Limit: depth grows exponentially in qubit count Q
5. Experimental Protocol	Translation Calorimeter – falsifiable measurement of S_A
HAL. Hidden Axioms	$S_{\text{Total}} = S_{\text{Obs}} + S_{\text{Base}} + S_{\text{Arch}} + S_{\text{Alg}} + S_{\text{Trot}}$

1.3 Prior Art

This work builds on and extends several independent lines of inquiry. De Gosson (2012) first framed the conflict between geometry and algebra in quantum information using Gromov’s non-squeezing theorem, but stopped at the existence argument without quantification. The QNFO Research Collective’s “Qubit Delusion” series (QNFO Research Collective 2026e, 2026a, 2026c, 2026d, 2026b) provided the philosophical scaffolding. Our contribution is the quantitative framework: explicit derivations of S_A , Rosetta’s Constant R , and the formal categorical proof of non-exactness.

2. Categorical Foundations

2.1 Category $\mathcal{P}$: Physical Dynamics

$\mathcal{P}$ has as objects **symplectic manifolds** (M^{2n}, ω) . Morphisms are Hamiltonian flows $\varphi_t^H : M \rightarrow M$ generated by $H : M \rightarrow \mathbb{R}$, or unitarily via $U(t) = e^{-iHt/\hbar}$. For the transmon: phase space is the cylinder $M = S^1 \times \mathbb{R}$ with $H = 4E_C n^2 - E_J \cos(\phi)$ (Koch et al. 2007).

2.2 Category $\mathcal{C}$: Computational Logic

$\mathcal{C}$ has as objects finite-dimensional state spaces $\mathcal{H}_2^{\otimes n}$. Morphisms are finite gate compositions from $\mathcal{G} = \{H, T, \text{CNOT}\}$. Objects are finite and discrete – no continuous coordinate, no symplectic form.

2.3 Theorem 1: Non-Exactness

Theorem 1. $F : \mathcal{P} \rightarrow \mathcal{C}$ is not exact – there exist invariants preserved in $\mathcal{P}$ with no image in $\mathcal{C}$.

Proof. By Gromov’s non-squeezing theorem (1985): the symplectic capacity $c_G(M, \omega)$ is invariant under symplectomorphisms. Under F , this maps to a finite discrete point set with zero capacity. An invariant in the source with no image in the target – the defining signature of a non-exact functor. $\square$

3. Algebraization Complexity

The transmon Hamiltonian contains $\cos(\phi)$ – a transcendental function. For computation it must be approximated by a finite polynomial.

Definition. The **Algebraization Complexity** $A(H, \varepsilon)$ is the minimum polynomial degree N preserving eigenvalues to within ε .

Theorem 2. For the transmon with $E_J/E_C \gg 1$:

$$A(H, \varepsilon) \approx \sqrt{\frac{8E_J}{E_C}} \cdot \frac{1}{2} \ln \left(\frac{1}{2\varepsilon} \right)$$

For $E_J/E_C = 50$, $\alpha_r = 1.9\%$, $\varepsilon = 10^{-4}$: $A \approx 224$.

Result: The transmon’s exponential charge-noise immunity $\exp(-\sqrt{8E_J/E_C})$ is mathematically identical to its algebraic intractability – a **Coherence-Algebraization Conservation Law**.

4. Thermodynamics of Translation

4.1 Approximation Entropy S_A

The Kullback-Leibler divergence between the full bosonic density operator and its discretized projection:

$$S_A = D_{\text{KL}}(\rho_{\text{full}} \parallel \rho_{\text{grid}}) = \ln \left(\frac{N_{\text{max}}}{d_{\text{comp}}} \right)$$

For the transmon ($N_{\text{max}} \approx 12$, $d_{\text{comp}} = 2$): $S_A \approx 1.79$ nats ≈ 2.58 bits per state projection.

4.2 Rosetta’s Constant R

$$R \equiv k_B T \cdot S_A = k_B T \cdot \ln \left(\frac{N_{\text{max}}}{d_{\text{comp}}} \right)$$

At $T = 15$ mK: $R \approx 3.7 \times 10^{-25}$ J per state projection. Per logical gate: $R_{\text{gate}} \approx 1.5 \times 10^{-24}$ J/gate. With all five entropy sources (§7): $Q_{\text{total}} \approx 1.4 \times 10^{-24}$ J/gate.

4.3 Landauer Extension

Landauer (1961) established $k_B T \ln 2$ for erasing one bit within the SAME domain. We extend to inter-domain:

$$Q_{\text{total}} = k_B T \ln 2 + k_B T \ln(N_{\text{max}}/2) = k_B T \ln(N_{\text{max}})$$

For the transmon: $Q_{\text{total}} \approx 2.48 k_B T$ – the Rosetta term is $3.6\times$ the Landauer term.

5. The Digital Speed Limit

5.1 Trotter Error

The standard digital simulation uses: $e^{-i(H_A+H_B)t} \approx (e^{-iH_A t/n} e^{-iH_B t/n})^n$. Error scales as $\|[H_A, H_B]\|$. For the transmon: $\|[H_{\text{harm}}, H_{\text{nonlin}}]\| \propto \hbar \omega_p \alpha_r$.

5.2 Trotter Wall

For fault-tolerant threshold $\varepsilon = 10^{-4}$:

$$\frac{G_{\text{effective}}}{G_{\text{advertised}}} \approx \frac{\omega_p}{\alpha_r} \approx 50$$

A “100-gate” circuit executes 5,000 sub-gates. True gate budget at $T_2^* \approx 100$ μs :

$$G_{\text{budget}} \approx \frac{T_2^*}{t_g \cdot (\omega_p/\alpha_r)} \approx 100$$

5.3 Fractal Boundary

Theorem 4. There exists $\alpha^* \approx 3\%$ below which weakly anharmonic bosonic arrays are classically simulable. Standard transmons at $\alpha_r \approx 1.9\%$ are BELOW this boundary.

6. The Translation Calorimeter

Two experiments on the same transmon at the same temperature:

Experiment	Method	Predicted Heat
A (Analog)	Natural evolution under H_0	$Q_A \approx 0$
B (Digital)	Trotterized gate sequence	$Q_B \approx k_B T \cdot S_A$

Falsification: If $\Delta Q = Q_B - Q_A = 0$ (within noise), the framework is refuted.

Feasibility: NIS junction calorimeter with 1 K sensitivity. $Q_{\text{single}} \approx 8.2 \times 10^{-25}$ J; 10^6 cycles yield $\Delta T \approx 0.82$ mK (measurable). 37 gates for 5σ (theory). Estimated cost: \$520K, 18–24 months.

7. Hidden Axioms Layer (HAL)

$$S_{\text{Total}} = S_{\text{Obs}} + S_{\text{Base}} + S_{\text{Arch}} + S_{\text{Alg}} + S_{\text{Trot}}$$

Term	Origin	Transmon Value (nats)
S_{Obs}	Human cognitive filter (binary thresholding)	1.79
S_{Base}	Radix mismatch (binary vs. native spectrum)	1.79
S_{Arch}	Archimedean time-slicing (uniform Trotter)	0.5
S_{Alg}	Algebraization (cosine $\rightarrow$ polynomial)	1.0
S_{Trot}	Commutator error ($[H_{\text{harm}}, H_{\text{nonlin}}] \neq 0$)	1.5
Total		6.6

Anthropocentricity: The qubit encoding is a human cognitive artifact. A Platonic observer with access to all observables would have $S_{\text{Obs}} = 0$ – the transmon is a “bad qubit” only relative to binary cognition.

Radix Contingency: Binary is the worst radix for representing real numbers. Ternary encoding reduces S_{Base} by 60%.

Continuum Fallacy: The Trotter error IS the Archimedean cost. Logarithmic time-stepping reduces sub-gates by ~50% with no hardware modifications.

8. Discussion

The transmon is not a “bad qubit.” It is succeeding at being a bosonic computer. Three paths forward:

1. **Embrace analog:** Use transmons for analog quantum simulation of bosonic Hamiltonians
2. **Switch to fermions:** Spins and ions have native discrete spectra – the functor F may be exact for them
3. **Build hybrid architectures:** Bosonic coprocessor + fermionic digital core, with bounded S_A at the interface

Falsifiability

This framework makes specific, testable predictions: 1. $\Delta Q > 0$ in calorimetry 2. $\Delta Q \propto -\ln(\alpha_r)$ across transmon variants 3. $\alpha^* \approx 3\%$ Fractal Boundary 4. Trotter Wall $\approx 50\times$ overhead 5. Logarithmic time-stepping yields 50% Trotter reduction

If $\Delta Q = 0$ – if digital translation is exact – the entire framework collapses. This is the hallmark of good science.

9. Conclusion

Digital computation is an engineering triumph but not a universal ontology. The universe runs on complex amplitudes and continuous phase, not binary. The Approximation Entropy S_A quantifies the distance between computational control and physical understanding – in Joules.

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